

Lecture 10: Completeness.

A statistic T is said to be complete, if for any real valued function $h(\cdot)$, $E_{\theta}[h(T)] = 0 \quad \forall \theta \in \Theta$, implies $h(t) = 0$ almost everywhere (a.e.)
(w, $h(T) = 0$ with probability 1).

2. Direct Approaches to investigate completeness of a statistic.

Verify if $E_{\theta}[h(T)] = 0 \quad \forall \theta \in \Theta$ implies $h(T) = 0$ w.p.1 or not

Integral transformations & their uniqueness property to establish completeness of a statistic.

1. Unilateral Laplace transform of a continuous f.p. $f(x)$

$$\phi_U(z) = \int_0^{\infty} e^{-zx} f(x) dx$$

2. Bilateral Laplace transform of a continuous f.p. $f(x)$

$$\phi_B(z) = \int_{-\infty}^{\infty} e^{-zx} f(x) dx$$

3. Mellin's transform of a continuous f.p. $f(x)$

$$\phi_M(z) = \int_0^{\infty} x^{z-1} f(x) dx$$

For two continuous f.p.s $f(x) + g(x)$, let $\phi_f(z)$ and $\phi_g(z)$ be their integral transforms, respectively.

Then $\phi_f(z) = \phi_g(z) \Rightarrow f(x) = g(x)$ a.e. (• indicates the 3 types of transforms w.r.t)

So, if $\phi_f(z) = 0$, then $f(x) = 0$ a.e.

Ex 1 r.v. $X \sim N(0, \theta)$, $\theta > 0$

Is $T = X^2$, the minimal sufficient statistic, complete? (Application of Mellin's transform & its uniqueness property)

$\therefore \frac{X}{\sqrt{\theta}} \sim N(0, 1)$, $\therefore \frac{X^2}{\theta} \sim \chi_1^2 \stackrel{d}{=} Y\left(\frac{T}{\theta}\right) \sim G\left(\frac{1}{2}, \frac{1}{2}\right)$, so that

pdf of Y is $f_Y(y) = \frac{1}{\Gamma(\frac{1}{2}) 2^{1/2}} e^{-y/2} y^{-1/2}$, $y > 0$

then pdf of T $f_T(t) = \frac{1}{\sqrt{2\pi} \sqrt{t\theta}} e^{-t/2\theta}$, $t > 0$; $\theta > 0$.

and $E_{\theta}[h(T)] = \int_0^{\infty} h(t) f_T(t) dt = \int_0^{\infty} h(t) (2\pi\theta t)^{-1/2} e^{-t/2\theta} dt$

Thus, $E_{\theta}[h(T)] = 0 \quad \forall \theta \in (0, \infty) \Rightarrow \int_0^{\infty} h(t) e^{-t/2\theta} t^{-1/2} dt = 0 \quad \forall \theta \in (0, \infty)$
 $\Rightarrow \int_0^{\infty} h(t) e^{-t/2} dt = 0 \Rightarrow h(t) e^{-t/2} = 0 \quad \forall t \in (0, \infty)$
 $\Rightarrow h(t) = 0 \quad \forall t \in (0, \infty)$ (i.e., a.e.)

Ex 2

 $X_1, \dots, X_n$ r.s. from $N(\theta, 1)$, $\theta \in \mathbb{R}$.

2/10

 $T = \bar{X}$ is a MSS.Verify completeness of $\bar{X}$ by the unilateral Laplace Transform & its uniqueness property.

Ex 3

 $X_1, \dots, X_n$ r.s. from $N(\mu, \sigma^2)$, $\mu \in \mathbb{R}$, $\sigma^2 > 0$. $T = (\bar{X}, S^2)$ is j'tly sufficient for (μ, σ^2) .

To verify completeness, check if

$$E_{(\mu, \sigma^2)}[h(\bar{X}, S^2)] = 0 \quad \forall \quad \theta = (\mu, \sigma^2)' \in (-\infty, \infty) \times (0, \infty)$$

implies $h(\bar{x}, s^2) = 0$ a.e., or not.(Use independence of $\bar{X}$ & S^2 when considering the j't density to find $E(h(\bar{X}, S^2))$.Further, note that marginally, $\bar{X} \sim N(\mu, \sigma^2/n)$

$$(n-1)S^2/\sigma^2 \sim \chi_{n-1}^2$$

Need the bilateral integral transform in this case.

Ex 4 ~~Direct by definition~~, $X_1, \dots, X_n$ iid $\text{Ber}(\theta)$, $\theta \in (0, 1)$. $T = \sum_{i=1}^n X_i$ is a MSS and $T \sim \text{Bin}(n, \theta)$.Check completeness of T (directly, by definition)

$$E_{\theta}(h(T)) = 0 \quad \forall \theta \in (0, 1) \Rightarrow \sum_{t=0}^n h(t) \binom{n}{t} \theta^t (1-\theta)^{n-t} = 0 \quad \forall \theta \in (0, 1).$$

$$\text{or, } \sum_{t=0}^n h(t) \binom{n}{t} \left(\frac{\theta}{1-\theta}\right)^t = 0 \quad \forall \theta \in (0, 1)$$

$$\text{or, } \sum_{t=0}^n h(t) \binom{n}{t} \lambda^t = 0 \quad \forall \lambda \in (0, \infty)$$

(a power series expansion in λ , which is equal to 0 $\forall \lambda \in (0, \infty)$).

$$\Rightarrow h(t) \binom{n}{t} = 0 \quad \forall t = 0, 1, \dots, n \quad (\text{each coeff. should be } \neq 0)$$

$$\Rightarrow h(t) = 0 \quad \text{a.e.}$$

Ex 5.

 $X_1, \dots, X_n$ iid $U(0, \theta)$, $\theta > 0$ $T = X_{(n)}$ is a MSS. Is it complete?To check if $E_{\theta}(h(T)) = 0 \quad \forall \theta > 0 \Rightarrow h(t) = 0$ a.e.

$$f_T(t) = \frac{n}{\theta^n} t^{n-1}, \quad 0 < t < \theta \quad \text{is the pdf of } T$$

$$E_{\theta}(h(T)) = \frac{n}{\theta^n} \int_0^{\theta} h(t) t^{n-1} dt = 0 \quad \forall \theta \in (0, \infty)$$

$$\Rightarrow \int_0^{\theta} h(t) t^{n-1} dt = 0 \quad \forall \theta \in (0, \infty) \quad (\text{can't apply any integral transform})$$

Differentiating both sides wrt θ ,

$$h(\theta) \theta^{n-1} - h(0) \cdot 0 = 0 \quad \forall \theta \in (0, \infty)$$

$$\Rightarrow h(\theta) = 0 \quad \forall \theta \in (0, \infty), \text{ i.e., } h(t) = 0 \quad \forall t \in (0, \theta) \cup \theta \in (0, \infty)$$

$$\Rightarrow h(t) = 0 \quad \text{a.e.}$$